Simulating Chemotactic Cluster Separation in the Keller-Segel Model: The Impact of Finite Domains and Spatial Discretization

Min Ki, Hong
(Dated: October 10, 2026)

Collective behaviour and pattern formation are fundamental features of biological systems mathematically described by the Keller-Segel model. While the standard Keller-Segel model provides robust theoretical predictions in the continuum limit, biological systems and numerical simulations are inherently composed of discrete agents operating within finite spatial domains. This report investigates the formation of chemotactic clusters through a hybrid computational simulation of the Keller-Segel model. The primary objective is to evaluate the simulation data against theoretical predictions derived from linear stability analysis, specifically verifying the theoretical proportionality between the dominant cluster separation wavelength ($\lambda_{\max}$) and the square root of the chemical diffusion coefficient ($\sqrt{D_{\text{chem}}}$). The simulation reproduced the expected scaling behaviour, yielding an empirical gradient of 0.67 ± 0.06 and a positive y-intercept offset of 2.61 ± 0.30 for the specific configuration of the simulation. The analysis evaluates the intrinsic limitations of modelling continuous spatial instabilities in a finite, discretised domain, demonstrating that the offset of accurately reflects the intrinsic discretisation limits of the simulation domain. Furthermore, it demonstrates how periodic boundary conditions introduce structural artifacts into the power spectrum, which can significantly alter the macroscopic behaviours predicted by continuous linear stability analysis.

1. INTRODUCTION

Collective behaviour and pattern formation are fundamental features of biological systems. A classic example of such self-organization is observed in the slime mould *Dictyostelium discoideum*, which aggregates into multicellular structures through chemotaxis, guided by self-generated gradients of diffusible chemoattractants such as cyclic AMP (cAMP) [1]. It is mathematically described by the Keller-Segel model as the foundational continuum framework [2]. It uses a system of coupled partial differential equations to capture the feedback loop between cell migration and chemical concentration dynamics.

While the standard Keller-Segel model provides robust theoretical predictions in the continuum limit, biological systems are inherently composed of discrete agents operating within finite spatial domains. Furthermore, translating this continuum framework into numerical simulations necessitates spatial discretization in finite spatial domain. These introduce local stochasticity and boundary constraints, both of which can significantly alter the macroscopic behaviors predicted by continuous linear stability analysis.

This report investigates the formation of chemotactic clusters through a hybrid computational simulation of the Keller-Segel model, coupling discrete stochastic cell agents with a continuous chemical field. The primary objective is to evaluate the simulation data against theoretical predictions derived from linear stability analysis. Specifically, this study extracts the dominant cluster separation wavelength ($\lambda_{\max}$) to test its mathematical relationship with the chemical diffusion coefficient (D_{chem}), verifying the theoretical proportionality $\lambda_{\max} \propto \sqrt{D_{\text{chem}}}$.

2. THEORY

2.1. Keller-Segel Model

The governing equations of the system can be simplified for constant parameters from the standard Keller-Segel model

$$\frac{\partial c}{\partial t} = D_{\text{chem}} \nabla^2 c - \gamma c + \alpha \rho \tag{1}$$

$$\frac{\partial \rho}{\partial t} = D_{\text{cell}} \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla c) \tag{2}$$

where each equation is a time-dependent partial differential equation (PDE) for the chemical concentration c and continuous cell population density ρ [2, 3]. Here, D represents the specific diffusion constant for each species, γ is the chemical degradation rate, α is the chemical production rate by cells, and χ is the chemotactic sensitivity.

While Equation 2 serves theoretical analysis, the simulation employs the overdamped Langevin equation

$$\vec{r}(t + \Delta t) = \vec{r}(t) + \chi \nabla c(\vec{r}) \Delta t + \sqrt{2 D_r \Delta t} \vec{\eta}. \tag{3}$$

which is the equivalent stochastic formulation for discrete agents [4, 5]. Here, $\vec{\eta}$ is the noise vector and D_r is equivalent to D_{cell}.

2.2. Linear Stability Analysis

Macroscopic features of the system can be theoretically predicted from the microscopic governing equations in the linear regime [6]. This involves assuming linear perturbations on the cell population and chemical field

$$\rho(\vec{r}, t) = \rho_0 + \delta\rho, \quad c(\vec{r}, t) = c_0 + \delta c \tag{4}$$

A quasi-steady state is also assumed, where the chemical diffuses much faster than the cells ($\frac{\partial c}{\partial t} \approx 0$). Ignoring secondary terms yields two linearized PDEs [7]

$$0 = D_{\text{chem}} \nabla^2 \delta c - \gamma \delta c + \alpha \delta \rho \tag{5}$$

$$\frac{\partial \delta\rho}{\partial t} = D_{\text{cell}} \nabla^2 \delta\rho - \chi \rho_0 \nabla^2 \delta c \tag{6}$$

Standard methods for solving linear PDEs provide the time-domain solution

$$\delta\tilde{\rho}(k, t) = \delta\tilde{\rho}_i(k) e^{\omega(k) t} \tag{7}$$

TABLE I. Simulation configuration parameters.

Parameter	Symbol	Value
Time step	Δt	10^{-5}
Grid resolution	Δx	1.0
Domain dimensions	$L_x \times L_y$	32×32
Total cells	N	10 240 000
Chemical diffusion	D_{chem}	16.0 to 64.0
Chemical decay rate	γ	0.1
Chemical secretion rate	α	1.0
Chemotactic sensitivity	χ	2.0
Cell diffusion	D_r	0.5
Inflated time factor	M	300

where the tilde represents the Fourier transform and $\delta\rho_i(k)$ is the initial condition $\rho(t=0)$. The angular frequency ω is given by the dispersion relation for each k-mode

$$\omega(k) = -D_{\text{cell}}k^2 + \frac{\chi\rho_0\alpha k^2}{\gamma + D_{\text{chem}}k^2}. \tag{8}$$

Setting $\omega(k) > 0$ yields the collapse condition and the critical density

$$\rho_0 > \frac{D_{\text{cell}}(\gamma + D_{\text{chem}}k^2)}{\chi\alpha} \implies \rho_c = \frac{D_{\text{cell}}\gamma}{\chi\alpha}. \tag{9}$$

Differentiating the dispersion relation allows the fastest growing mode to be identified as

$$\lambda_{\max} = 2\pi\sqrt{\frac{D_{\text{chem}}}{\gamma}}\left(\sqrt{\frac{\rho_0}{\rho_c}} - 1\right)^{-1/2} \tag{10}$$

which demonstrates the proportionality $\lambda_{\max} \propto \sqrt{D_{\text{chem}}}$.

3. METHODOLOGY

3.1. Simulation

The system is modeled using a hybrid framework. Individual cells are tracked as discrete agents, and their stochastic movement is integrated via the Euler-Maruyama method (Equation 3). The continuous chemoattractant field $c(\vec{r}, t)$ is solved using a forward-time central-space (FTCS) finite-difference scheme (Equation 1). To efficiently maintain the chemical field in a quasi-steady state, an inflated time multiplier M is applied to the chemical evolution equation, as detailed further in Section 3.2.

Simulations are conducted in a two-dimensional domain ($L_x \times L_y$) with periodic boundary conditions. At $t = 0$, N cells are distributed uniformly at random. To accelerate convergence, the initial chemical field is set uniformly to its theoretical steady-state concentration, $c_0 = \frac{\alpha\rho_0}{\gamma}$, where ρ_0 is the mean cell density.

The total number of cells N is chosen to ensure $\rho_0 > 1000$ cells per grid square. This limits the local Poisson noise to below 3%, preventing large stochastic fluctuations from disrupting the linear regime. To guarantee that the system remains strictly within this linear regime during observation, all analyses are performed exactly when the maximum local density reaches $1.1\rho_0$. The total simulation time is adjusted accordingly for each run to capture this specific point.

3.2. Quasi-Steady State Analysis

Without the quasi-steady state assumption for Equation 1, a homogeneous equation for the chemical field is obtained as $\delta c = Ce^{-\Gamma_c t}$, where $\Gamma_c = D_{\text{chem}}k^2 + \gamma$ is the relaxation rate.

Satisfying the quasi-steady assumption requires $\Gamma_c \gg \omega$. The maximum ratio ω/Γ_c under the current configuration is 20 at $D_{\text{chem}} = 16$, indicating that cells cluster much faster than the chemical field relaxes.

To satisfy the quasi-steady assumption, the relaxation rate of the chemical field must be artificially increased. This is achieved by setting an inflated time step $dt_{\text{chem}} = M \times dt$ with a multiplier of $M = 300$. Numerical stability must be verified via the Von Neumann limit

$$D_{\text{chem}}\frac{dt_{\text{chem}}}{dx^2} = 64 \times \frac{0.00001 \times 300}{1^2} = 0.192 \leq 0.25. \tag{11}$$

This confirms that the inflated time step satisfies the numerical stability limit while successfully decoupling the timescales.

3.3. Extracting k_{max} via Maximum Likelihood Estimation

To accurately extract the dominant wavenumber $k_{\max}$, Maximum Likelihood Estimation (MLE) is employed rather than standard least-squares fitting. Assuming translational invariance in the system, the real and imaginary components X and Y are independent. By the Central Limit Theorem, they can be approximated as normally distributed variables with zero mean and variance $\sigma^2 = S(k)/2$, where $S(k) = \langle|Z(\vec{k})|^2\rangle$ is the expected underlying power spectrum. Consequently, the measured power $P = X^2 + Y^2$ at a given wavenumber is the sum of two squared Gaussian variables, following an exponential distribution

$$f(P|S(k)) = \frac{1}{S(k)}\exp\left(-\frac{P}{S(k)}\right). \tag{12}$$

Because this distribution is highly skewed and its most probable value is strictly $P = 0$ for any given k, applying standard least-squares regression introduces significant parameter bias due to the intrinsic assumption of symmetric Gaussian errors [8].

The expected power is modeled as a Gaussian peak on a constant white-noise background

$$S(k) = A\exp\left(-\frac{(k - k_{\max})^2}{2w^2}\right) + C, \tag{13}$$

where A, w, $k_{\max}$, and C are adjustable parameters. For a set of N independent discrete power spectrum measurements P_i at wavenumbers k_i, the log-likelihood function is given by

$$\ln\mathcal{L} \propto \sum_{i=1}^{N}\left[-\ln S(k_i) - \frac{P_i}{S(k_i)}\right] \tag{14}$$

Maximizing this log-likelihood across the parameter space yields an unbiased estimate for the dominant wavenumber $k_{\max}$.

The covariance matrix Σ is given by the inverse of the Fisher Information Matrix $\mathcal{I}$

$$\Sigma = \mathcal{I}^{-1}, \quad \text{where} \quad \mathcal{I}_{jk} = \sum_i \frac{1}{S_i^2}\frac{\partial S_i}{\partial\theta_j}\frac{\partial S_i}{\partial\theta_k} \tag{15}$$

where the Fisher information matrix is the expected value of the negative Hessian matrix of the specific log-likelihood function.

4. RESULTS

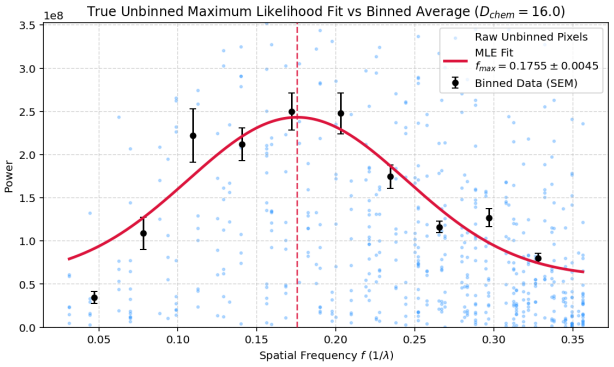


FIG. 1. True unbinned MLE fit versus binned average for $D_{\text{chem}} = 16.0$. Uncertainty visibly increases as the spatial frequency $f_{\max}$ approaches zero.

Figure 1 shows an example of the MLE fitting, confirming that the method successfully extracts the signal. This process was repeated across different D_{chem} values to extract the dominant wavelengths ($\lambda_{\max}$) and compare the data to the theoretical model.

Figure 2 shows that the simulation reproduced the expected scaling behavior of the cluster separation. The extracted $\lambda_{\max}$ has a clear linear relationship with the square root of the chemical diffusion coefficient ($\sqrt{D_{\text{chem}}}$). The empirical fit gives a gradient of $m_{\text{fit}} = 0.67 \pm 0.06$, which is 2.2σ away from the continuous theoretical value of $m_{\text{theory}} = 0.79$. The linear regression also shows a positive y-intercept offset of $C = 2.61 \pm 0.30$. This offset is discussed further in the Discussion section.

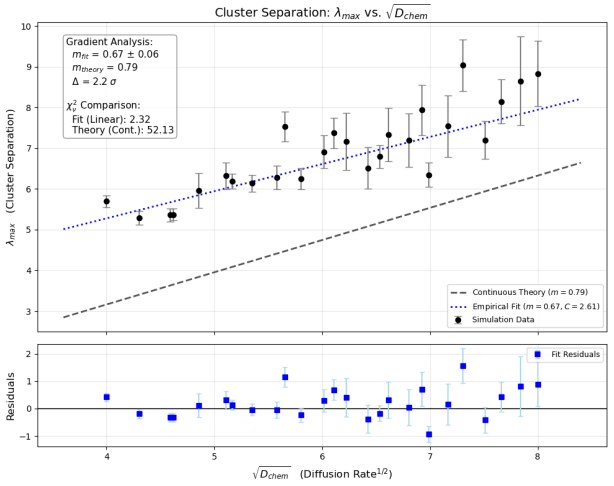


FIG. 2. Cluster separation $\lambda_{\max}$ vs. $\sqrt{D_{\text{chem}}}$. The empirical fit yields a gradient of $m = 0.67 \pm 0.06$ and an offset of $C = 2.61 \pm 0.30$. Residuals display an oscillatory pattern around the fit.

Although the data follows the expected linear trend, a reduced chi-squared value of $\chi^2_\nu > 2$ indicates the linear model does not fully capture the behavior. The residuals do not scatter randomly but instead oscillate around the fit, which is likely a fitting artifact caused by the finite grid size and periodic boundary conditions. Additionally, the uncertainty in $\lambda_{\max}$ increases noticeably at larger D_{chem} values because the symmetric Gaussian model for the expected power is no longer suitable. These issues are discussed further in Section 5.

5. DISCUSSION

The observed constant offset of $C = 2.61 \pm 0.30$ is a sensible physical consequence of the finite simulation grid. In a pixelated space, the system cannot support infinitely small wavelengths. The hard theoretical minimum is 2 pixels to represent a contiguous peak and trough horizontally or vertically, or approximately $\sqrt{2} \approx 1.4$ pixels diagonally. Thus, a baseline structural offset near 2.6 accurately reflects the intrinsic discretization limits of the simulation domain rather than a breakdown of the physical model.

The oscillatory behaviour observed in the residuals is a direct artifact of the periodic boundary conditions (PBC) applied to a discrete grid. Under PBCs, the system can only support discrete spatial modes defined by

$$\lambda = \frac{N_{\text{pixel}}}{\sqrt{n_x^2 + n_y^2}} \tag{16}$$

where $N_{\text{pixel}} = 32$ represents the total length of one side of the domain. Because the system cannot access a continuous spectrum of $\lambda_{\max}$, the resulting power spectrum frequently becomes skewed or exhibits double-peak distributions. Attempting to fit a symmetric Gaussian model to these discrete, non-Gaussian data points forces the extracted dominant wavelength to oscillate around the true continuous theory line.

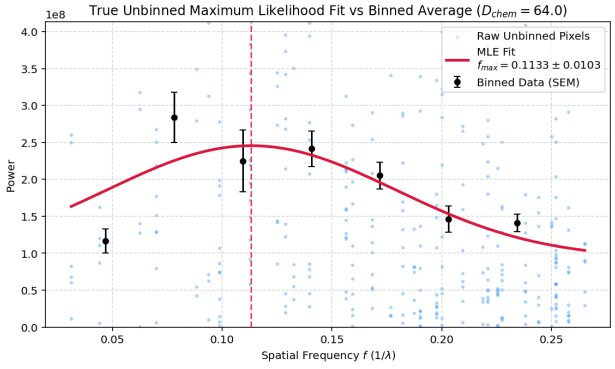


FIG. 3. MLE fitting on an asymmetric curve. The symmetric Gaussian model struggles to accurately capture the expected power spectrum as $f_{\max}$ approaches zero.

This artifact becomes more pronounced at larger cluster separations (higher λ, smaller f). As the dominant spatial frequency $f_{\max}$ shifts closer to zero, the symmetric Gaussian model for the expected power spectrum systematically struggles. Visual inspection of the asymmetric power spectrum at $D_{\text{chem}} = 64$ as shown in Figure 3 confirms that the true $f_{\max}$ is likely smaller than the value identified by a symmetric fit.

To properly resolve this boundary constraint in future models, the MLE procedure should employ an asymmetric distribution, such as a log-normal distri-

bution in frequency space:

$$S(f) = \frac{A}{f} \exp\left(-\frac{(\ln f - \mu)^2}{2w^2}\right) + C \qquad (17)$$

This function naturally accommodates the asymmetric nature of the power distribution near the zero-frequency bound, eliminating the parameter bias introduced by symmetric fitting models and likely resolving the oscillatory residuals. Still, this model is empirical model and has no theoretical justification.

6. CONCLUSION

This report investigated the formation of chemotactic clusters through a hybrid computational simulation of the Keller-Segel model. The simulation reproduced the expected scaling behaviour of the cluster separation, where $\lambda_{\max}$ has a clear linear relationship with the square root of the chemical diffusion coefficient. The empirical fit gives a gradient of 0.67 ± 0.06 and a positive y-intercept offset of 2.61 ± 0.3. The offset accurately reflects the intrinsic discretization limits of the simulation domain. Additionally, the oscillatory behaviour observed in the residuals is a direct artifact of the periodic boundary conditions applied to a discrete grid. Ultimately, this study demonstrates how spatial discretization and finite computational grids impose boundary constraints and structural artifacts that can significantly alter the macroscopic behaviours predicted by continuous linear stability analysis.

[1] T. M. Konijn, J. Van de Meene, J. T. Bonner, and D. S. Barkley, Proceedings of the National Academy of Sciences **58**, 1152 (1967).

[2] E. F. Keller and L. A. Segel, Journal of Theoretical Biology **26**, 399 (1970).

[3] J. D. Murray, *Mathematical Biology I: An Introduction*, 3rd ed., Interdisciplinary Applied Mathematics, Vol. 17 (Springer, New York, 2002).

[4] P. Romanczuk, M. Bär, W. Ebeling, B. Lindner, and L. Schimansky-Geier, The European Physical Journal Special Topics **202**, 1 (2012), arXiv:1202.2442 [cond-mat.other].

[5] P.-H. Chavanis, Communications in Nonlinear Science and Numerical Simulation **15**, 60 (2010).

[6] J. D. Murray, *Mathematical Biology II: Spatial Models and Biomedical Applications*, 3rd ed., Interdisciplinary Applied Mathematics, Vol. 18 (Springer, New York, 2003).

[7] W. Jäger and S. Luckhaus, Transactions of the American Mathematical Society **329**, 819 (1992).

[8] S. F. Nørrelykke and H. Flyvbjerg, Review of Scientific Instruments **81**, 075103 (2010).